

The Seed Vault Guide

Every step to save, dry, and store seeds from the vegetables you already grow

She bought one plant. Then she never bought it again. Here is the complete system — which vegetables to start with, how to harvest and dry correctly, and how to store seeds that remain viable for a decade.

BEST VEGETABLES TO START WITH

- **Tomatoes** — easiest. Seeds are enclosed in gel that requires a wet extraction step (see below). Open-pollinated and heirloom varieties only; hybrids do not breed true.
 - **Peppers** — almost as easy as tomatoes. Seeds are dry inside. Wait until the pepper is fully colored and slightly wrinkled.
 - **Beans and peas** — the simplest of all. Let the pods dry completely on the plant until papery and brown. Shell and store.
 - **Lettuce** — let one plant bolt (go to flower) and allow the seed heads to dry on the plant. Shake seeds into a bag.
 - **Squash and zucchini** — seeds are large and easy to handle. Wait until the fruit is fully mature — past eating stage, skin hardened.
 - **Avoid F1 hybrid seeds** — the label will say 'F1 hybrid.' Seeds from these plants will either not grow or produce inconsistent, often inferior offspring. Only save from open-pollinated or heirloom varieties.
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KNOWING WHEN THE FRUIT IS READY

- **The seed is only mature when the fruit is finished** — for eating vegetables, this means letting the fruit go well past peak harvest.
 - **Tomatoes** — slightly overripe, soft, deeply colored. Seeds inside should be cream to tan, not white.
 - **Peppers** — fully colored (red, orange, or final color), slightly wrinkled at stem, beginning to soften.
 - **Beans** — pods have turned yellow-brown and feel dry and papery. Seeds rattle inside.
 - **Squash** — skin is so hard it resists a fingernail. Left on the vine past the first light frost if possible.
 - **General rule** — if you would not leave it on the plant any longer, leave it one more week.
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WET EXTRACTION (TOMATOES, SQUASH)

- Cut the fruit and scoop seeds with their surrounding gel into a glass jar.
 - Add an equal amount of water. Stir.
 - **Leave uncovered at room temperature for 2–3 days. A layer of mold will form on the surface** — this is correct. The fermentation breaks down the gel coat that inhibits germination.
 - After 2–3 days, add more water and stir vigorously. Viable seeds sink; hollow seeds and debris float.
 - Pour off the floating material and debris. Strain the sunken seeds through a fine sieve.
 - Rinse under cold water until clean.
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DRY EXTRACTION (PEPPERS, BEANS, PEAS, LETTUCE)

- **Peppers** — slice open and pull or scrape seeds from the center. Lay flat on a paper plate.
 - **Beans and peas** — shell dry pods by hand. Seeds are ready immediately.
 - **Lettuce** — shake dry seed heads over a paper bag or bowl. The seeds fall freely.
 - **No water needed** — these seeds carry no germination-inhibiting gel coat.
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DRYING SEEDS CORRECTLY

- **Spread seeds in a single layer on a paper plate or unbleached parchment** — never plastic, which traps moisture.
 - **Dry in a warm, airy room for a minimum of two weeks. Do not rush with artificial heat** — temperatures above 35 °C (95 °F) damage germination capacity.
 - Stir once daily for the first week to prevent clumping.
 - **The dryness test** — pinch a seed firmly between your fingers. If it bends, it needs more drying. If it snaps or shatters cleanly, it is ready.
 - **A second test** — try to dent a hard seed (bean, squash) with your thumbnail. If it dents, it still holds moisture.
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STORAGE: PAPER ENVELOPE VS. GLASS JAR

- **Paper envelope** — breathable; good for short-term storage (1–2 years) in a cool, dry environment. Write species, variety, and year on the outside. Fold and seal with a small label.
 - **Glass jar with tight lid** — ideal for long-term storage. Add a small silica gel packet inside to absorb residual moisture. Label the outside with tape.
 - **Never store in plastic bags long-term** — they trap residual moisture and accelerate mold.
 - **Store all seeds in a cool, dark, dry place** — a drawer, cellar, or pantry away from the stove. Temperature swings are the enemy of viability.
 - **Refrigerator storage** — sealed glass jar with silica gel, inside a larger sealed container to prevent condensation when removing. This extends viability to the maximum possible.
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SEED VIABILITY BY VEGETABLE

- **Tomatoes** — 4–6 years when properly dried and stored.
 - **Peppers** — 2–3 years.
 - **Beans and peas** — 3–4 years.
 - **Lettuce** — 1–3 years (shorter viability than most; save extra).
 - **Squash and zucchini** — 4–6 years.
 - **Onions and leeks** — 1–2 years only. Save fresh every season.
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COMMON MISTAKES TO AVOID

- **Picking too early** — seeds from underripe fruit have low germination rates. The most common reason saved seeds fail.
 - **Storing damp** — a single damp seed can mold and spread to the others in a sealed container. The snap test saves you from this.
 - **Freezer myths** — frozen seeds can work, but only if bone-dry first and sealed airtight. One cycle of condensation from a poorly sealed container destroys them. Start with cool-dark-dry; add freezer only once you are confident in your drying process.
 - **Saving from hybrids** — seeds saved from F1 plants are genetically unpredictable. Always verify the variety before saving.
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KEEP THIS IN MIND

Label everything immediately after drying — never 'do it later.' Unlabeled envelopes of small seeds are indistinguishable from each other. Write the species, variety, and harvest year the day you package them.

Comment LAMP tomorrow — Sunday's reel covers six hours of light from twelve cents worth of kitchen supplies.

SOURCES Seed Savers Exchange — seed saving guides by vegetable (seedsavers.org) · University of Minnesota Extension — Saving Vegetable Seeds · Colorado State University Extension — Storing Vegetable and Flower Seeds · National Center for Home Food Preservation — seed storage guidelines

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